LESSON PLAN

SFNA (MOD AIRMANSHIP) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various types of fuels, their properties and usage.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various types of fuels, their properties and usage. usages of additives
5. Specific Objectives	24.1.3.1 Trainee will be able to identify various types of greases, fluids and compounds.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of fuel types and additives from samples; Safety: PPE mandatory; Supervision: Instructor monitors handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of aviation fuel?
(b) Name one common additive used in aviation fuels.
(c) Why is fuel quality critical for aircraft performance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Jet A and Jet A-1 specifications • Kerosene-based fuel properties • Freezing point and flash point • Water and sediment tolerance • Density and specific gravity	L & Ds	PPT, IB	15 min
(b)	• Fuel additives and their functions • Corrosion inhibitors usage • Static electricity dissipaters • Ice prevention agents • Metal passivators role	L & Ds	PPT, PDF	20 min
(c)	• Storage and handling procedures • Contamination prevention measures • Dispensing equipment operation • Safety protocols during transfer • Inspection and testing methods	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between Jet A and Jet A-1 regarding freezing points.
(b) Describe how corrosion inhibitors protect fuel system components.
(c) List three safety precautions when handling aviation fuels.

12. Home Assignment and Follow Up	Prepare a brief note on the specific properties required for high-altitude fuel usage to be discussed in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min